

you to examine the problems one after another and correct them.

Properties panel. Properties panel offers the possibility to edit all the important properties of your layout elements without calling special dialogues. With the help of multi-selection, you can even change properties of as many elements as you wish at once.

Gerber/Excellon export. Sprint-Layout produces Gerber files (RS-274X), which are used to exchange layout data with PCB manufacturers. Drillings can be exported in the common Excellon format. There are also many PCB manufacturers who accept Sprint-Layout files.

Isolation milling. Sprint-Layout supports isolation-milling, where the PCB is produced using a CNC-milling machine. Sprint-Layout is able to calculate all the necessary isolation channels. The output file is produced in HPGL format.

Layer administration. Sprint-Layout offers a copper and a silk-screen layer for both sides of the board. There are two optional inner layers (for multilayer PCBs) and an outline layer (for the outline of your board) available. The layers can be switched on and off whenever necessary.

High resolution and accuracy. Sprint-Layout offers an increased resolution and accuracy. It is possible to define a grid value up to 1 µm. So you can easily design even very fine layout structures.

Printing. Various options are available to print your layouts—solder masks, component-mounting plans, silk screens, and drilling plans on paper sheets or transparent film. It is possible to repeat the printout of small designs in X- and Y-direction, to save film and material.

FX-RTOS

FX-RTOS primarily serves the needs of embedded systems with hard real-time constraints. However, the system has a modular design that allows the OS to be used in various environments. You can also utilise

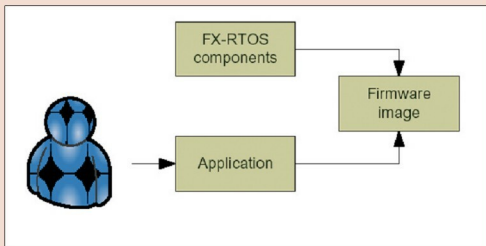


Fig. 7: FX-RTOS

some components of FX-RTOS for applications that don't need the whole operating system at all.

Highly configurable nature of FX-RTOS provides the ability to define multiple implementations for any system interface. A proper implementation for each component can be chosen during the configuration process. It is possible to build different operating systems with diverse properties and choose a proper system that matches the application needs as close as possible. For different configurations, even the whole system architecture may vary—for instance, almost any component may be excluded, or components responsibility or dependencies may be changed.

Two files appear as a result of the configuration and building process: a header file containing all the publicly available declarations, and a static library containing implementation of the OS functions. A user application should be statically linked against this library, as shown in Fig. 7.

Some important features of this embedded RTOS are:

1. Component architecture does not have mandatory elements. So unused components are not included in the target image
2. Support for symmetric multiprocessing
3. Use of simple pre-emptive event handlers without full-fledged threads
4. The operating system is con-

figurable using dependency injection technique

5. Reusable components can be combined in various configurations
6. Support for various scheduling policies (priority-based, FIFO, round-robin and their combinations)
7. Extensibility with user-defined components
8. Interrupt handling is quite effective
9. Easy cross-platform portability
10. POSIX support (optional)
11. Cross-platform utilities for configuring and building the OS
12. Provision of source code

TopoR

TopoR (Topological Router) is a unique high-performance topological router for PCB design.

TopoR combines the following characteristics:

1. Fast routing with high quality, making it even better than time-consuming manual routing
2. Availability of a set of powerful tools to shorten the design time of electronic components
3. Unique algorithms and alternative approach for solution of complex problems

TopoR reduces PCB design cycles, increases their technological effectiveness, reliability and quality, and lowers production costs.

Reducing the number of vias and layers, as well as the board size is the key to reduce PCB produc-